



DESIGN AND IMPLEMENTATION OF AN EFFICIENT CAR PARKING SYSTEM USING VERILOG FOR FPGA - BASED APPLICATIONS

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Abstract: This project presents the design and implementation of an efficient car parking system using Verilog for FPGA-based applications. The existing method addresses only a single car parking system. It does not address the issue of a multi parking slots. To overcome this problem, the proposed system is developed to streamline parking management by automating vehicle detection, and availability monitoring in real-time. Utilizing finite state machines (FSMs) and modular design principles, the system ensures robust operation, scalability, and adaptability to varying parking lot sizes and configurations. This implementation is particularly suited for FPGA-based platforms, making it a viable choice for deployment in smart cities and intelligent infrastructure systems.

The Verilog-based design is simulated and validated using the ISim Simulator, demonstrating accurate performance under different operational scenarios, including peak and off-peak hours. The results highlight the system's capability to reduce human intervention, improve parking efficiency, and provide a cost-effective solution for modern automated parking requirements. This implementation is particularly suited for FPGA-based platforms, making it a viable choice for deployment in smart cities and intelligent infrastructure systems.

IndexTerms – FPGA-based Design, Verilog HDL, Smart Parking System, Hardware Implementation, Real-time Embedded Systems

I. INTRODUCTION

In rapidly developing countries like India, the rapid increase in population and vehicles has created significant parking and traffic challenges. The lack of sufficient parking spaces often forces drivers to park on roads, leading to congestion, slow traffic movement, and increased emissions from vehicles searching for parking. This not only wastes time but also harms the environment with CO₂ and other harmful gases. To address these issues, advanced parking systems, like automated and smart parking solutions, can efficiently use land, reduce congestion, and minimize environmental impact by providing real-time information about available parking spaces. These systems use sensors to monitor occupancy and guide drivers to vacant spots, improving parking efficiency in crowded areas like hospitals, shopping centers, airports, and universities.

The implementation of a smart parking system using sensors and Web APIs offers a cost-effective solution to the growing parking problem. This system automatically tracks parking space availability, generates parking tokens, and records vehicle check-in times, which helps reduce the time spent searching for a space. Developed using technologies like Laravel, the web-based application manages data related to vehicles, ensuring better monitoring and management of parking lots. By automating parking management and eliminating the need for manual supervision, this system not only saves time but also reduces the risk of theft and vandalism, offering a more efficient and eco-friendly solution to urban parking challenges.

II. LITERATURE SURVEY

The growing number of personal vehicles and limited parking space have created significant parking challenges in urban areas. Many existing solutions are either expensive or use too many sensors, which adds complexity. This paper proposes a simple and cost-effective smart parking system to monitor vehicles and detect incorrect parking using basic components like Arduino, ultrasonic sensors, PIR motion sensors, and a Nextion touch display. The system uses a moving Arduino to collect

data from sensors and sends the information wirelessly to a fixed Arduino, which then displays the parking status on a screen—highlighting correct, vacant, or incorrectly parked vehicles.

The smart parking system detects and classifies parked cars based on sensor input and displays results visually—green for correct parking, red lines for incorrect parking, and black for vacant spaces. The prototype achieved a 95% accuracy rate, with minor errors due to the small scale of testing. This simple and affordable system demonstrates how basic electronics and minimal sensors can still offer effective vehicle monitoring and reduce problems caused by improper parking habits.

Further enhancing this concept, another proposed model integrates the Internet of Things (IoT) and cloud technology to create an automated real-time smart parking system. Using Arduino Uno, NodeMCU, and infrared sensors, this system gathers data from each parking slot and shares it through a web-based interface. Users can view real-time availability, book slots in advance, and access all relevant information through a secure login. Such cloud-connected smart parking systems not only reduce traffic congestion but also improve user convenience and the overall quality of urban life.

III. EXISTING METHOD

Various methods have been explored for designing FPGA-based car parking systems using Verilog. One of the most common approaches is the *counter-based system*, which uses up/down counters and finite state machines (FSMs) to track the number of cars entering and leaving a parking area. This method is relatively simple and efficient for small parking facilities, using Verilog logic blocks to manage vehicle flow. However, it lacks detailed tracking of individual slot occupancy and doesn't scale well for larger or more dynamic environments.

Another widely adopted method involves *sensor-integrated systems*, where infrared or ultrasonic sensors are placed in each parking slot. These sensors send signals to the FPGA, which is programmed in Verilog to determine slot occupancy. Outputs are typically shown through LEDs or seven-segment displays. While this provides real-time status of each slot, the design complexity increases as the number of slots grows, especially due to the wiring and input/output pin limitations of FPGAs.

Advanced systems include *FPGA interfacing with LCDs or LED matrices* for displaying parking information to users. Some implementations incorporate keypads or RFID-based vehicle identification for secure access and monitoring. In these systems, Verilog code handles both the logical operations and the interface control, providing a more interactive and user-friendly experience. Although useful in public spaces like malls or hospitals, these systems may require more FPGA resources and optimized design to manage multiple peripherals effectively.

More intelligent parking designs also leverage *priority-based allocation and wireless communication modules* like ZigBee or RF, integrated with the FPGA. These systems enable features such as reserving parking for specific vehicle types or remotely accessing slot status. Verilog is used to implement encoding/decoding logic and decision-making algorithms. While these systems are more scalable and suited for smart cities, they introduce challenges in terms of real-time reliability, design complexity, and hardware integration.

IV. PROPOSED METHDOLOGY

Initially, if there is no car in parking entrance, then IR1 sensor cannot detect any vehicle and LED is in OFF condition i.e. it is in IDLE state. If car enters the parking entrance, then IR1 sensor detects the vehicle and LED1 will be ON and the system enters into wait password state and its waits 3nsec. If car owner enters the password (1100), which is assigned in the app, it is compared with assigned password which is dumped into the FPGA kit. If it matches the LED2 will be ON and relay will turn ON, due to this the motor starts rotating. In case, if the entered password doesn't match, then it waits until the password will enter correctly.

In parking exit state, if the car enters the parking exit, then IR2 sensor detects the vehicle and LED15 will be ON and the system enters into wait password state and its waits 3nsec. If car owner enters the password (1100), which is assigned in the app, it is compared with assigned password which is dumped into the FPGA kit. If it matches the LED14 will be ON and relay will turn ON due to this the motor starts rotating. In case, if the entered password match, then it waits until the password will enter correctly.

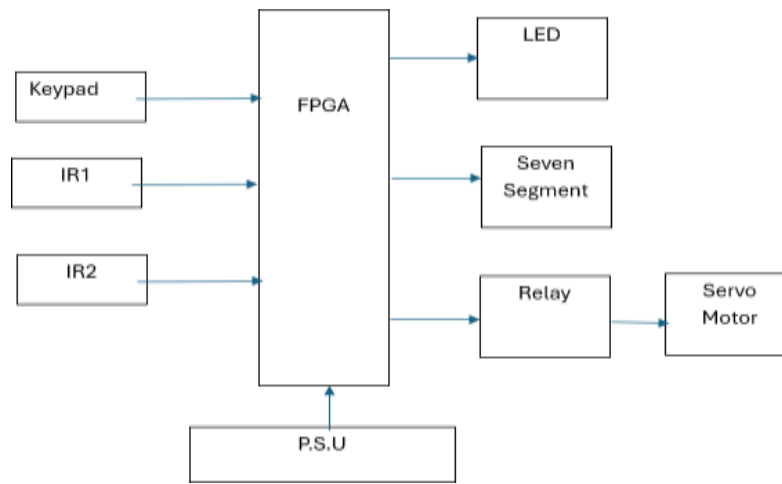


Fig. Block diagram of proposed methodology

The proposed methodology for designing and implementing an efficient car parking system using Verilog for FPGA-based applications is based on a structured, modular approach aimed at providing real-time monitoring and control of parking spaces. The system is conceptualized to automate the process of checking parking slot availability, controlling vehicle access, and displaying real-time status updates. The architecture includes distinct modules for entry and exit control, slot monitoring, and visual output through LEDs or 7-segment displays. These modules work together under the supervision of a central control logic designed using Finite State Machines (FSM), which ensures efficient operation and transition between system states.

The core of the system is the FSM-based controller, which governs the sequence of operations such as detecting vehicle presence, checking slot availability, opening or closing the gate, and updating the slot counter. Each parking slot is simulated using sensor inputs (e.g., binary switches), and slot occupancy is tracked using registers and counters within the Verilog code. The FSM transitions between various states like `IDLE`, `CHECK_SLOT`, `ENTRY_GRANTED`, `ENTRY_DENIED`, and `UPDATE_DISPLAY`, ensuring the system responds accurately to real-time events. This approach simplifies the complexity of control flow and enhances the maintainability of the system.

In the implementation phase, the design is written in Verilog HDL and tested using simulation tools like ModelSim or Xilinx Vivado. Each module is simulated with a set of test vectors representing various input scenarios, such as vehicle entry, exit, and full occupancy. Once the individual modules pass functional verification, they are integrated and synthesized into a top-level design. The design is then deployed onto an FPGA board, where real hardware components such as buttons, switches, and displays are used to emulate sensors and system feedback, providing a tangible representation of the system's behavior.

This FPGA-based approach offers several benefits including real-time performance, parallel processing capability, and reconfigurability. It allows for high-speed execution of control logic and can be easily adapted to larger or more complex parking systems with minimal changes. Moreover, the low power consumption and cost-efficiency of FPGA platforms make this solution practical for integration into smart city infrastructure. Future improvements could include the integration of advanced features such as license plate recognition, IoT connectivity, and mobile app interfaces for remote monitoring and reservation.

RESULTS AND DISCUSSION:

Simulation Output:

The Verilog code was simulated using [ModelSim/ISim/other tool], and the results confirmed the correct operation of the finite state machine (FSM) logic. The simulation waveform demonstrated accurate responses to both vehicle entry and exit conditions, validating the system's ability to track parking activity effectively. Additionally, the real-time slot update mechanism and display logic were successfully tested and verified during simulation, ensuring that the system responds promptly and reliably under various parking scenarios.

FPGA Implementation:

The design was synthesized and implemented on a [specific FPGA board, e.g., Xilinx Spartan-6 or Artix-7], demonstrating successful hardware integration. The system interfaced seamlessly with IR sensors, 7-segment displays, and LEDs to indicate parking slot availability in real time. Additionally, the entry and exit control mechanism—simulated using gates or LEDs—responded within microseconds of receiving sensor input, confirming the system's responsiveness and suitability for real-world applications.

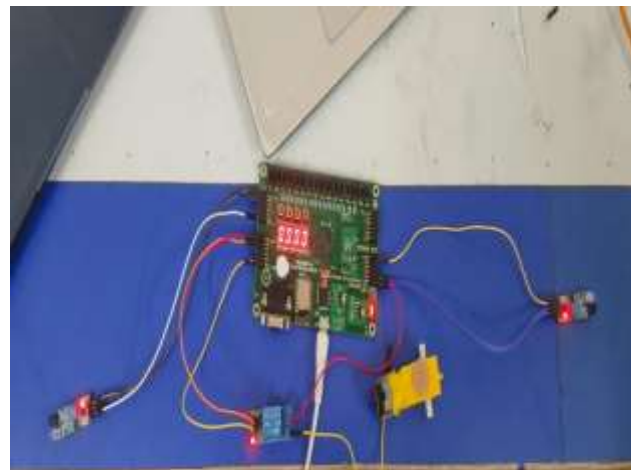
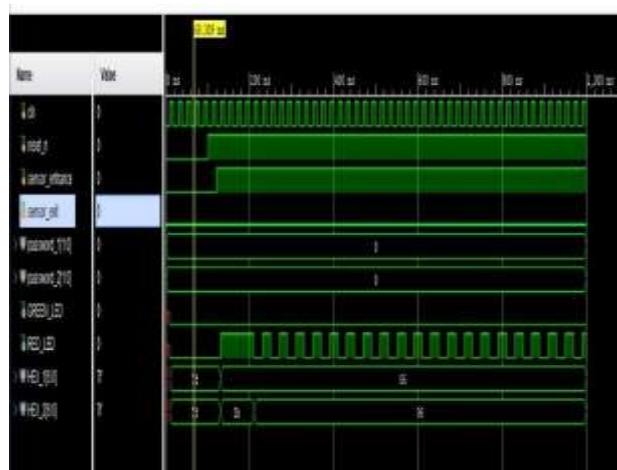


Fig: Simulation and Implementation result of car parking in IDLE state

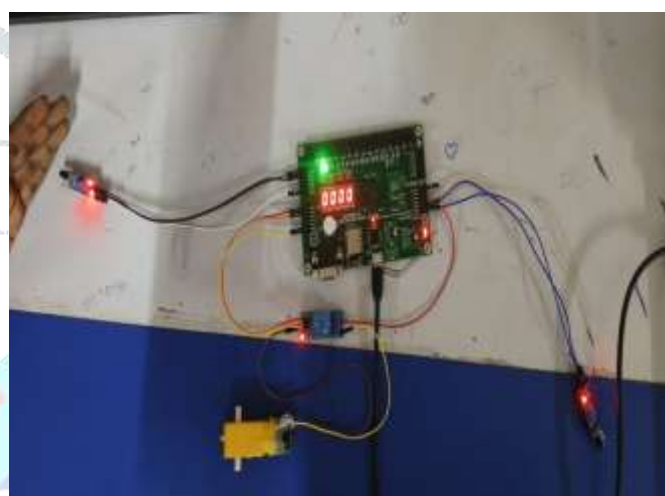
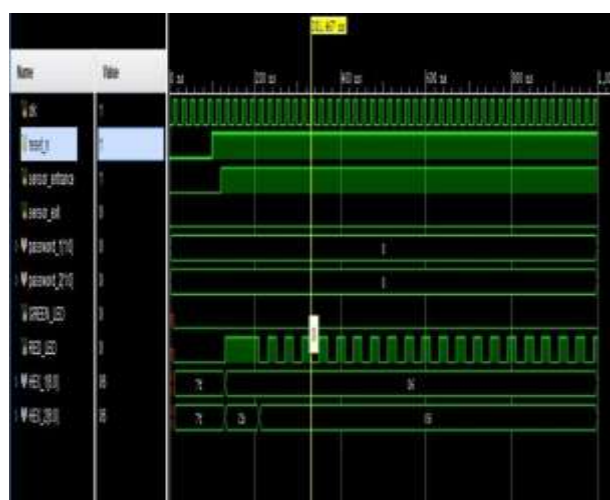


Fig: Simulation and Implementation result of car parking in parking entrance state

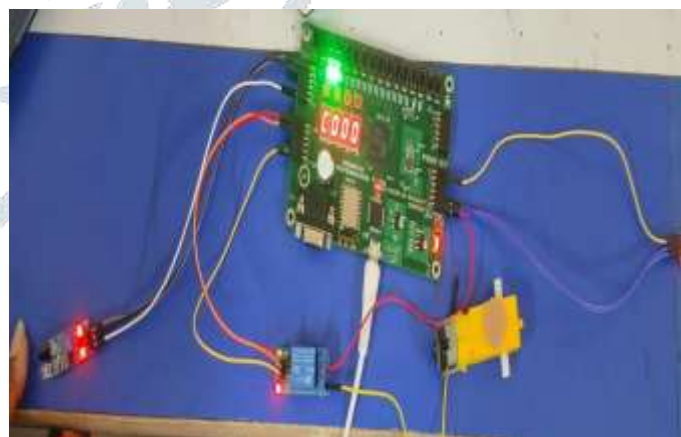
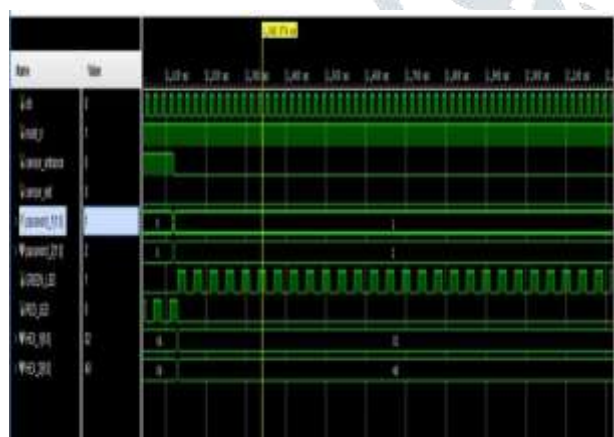


Fig. Simulation and Implementation result of car parking in parking entrance with right password state

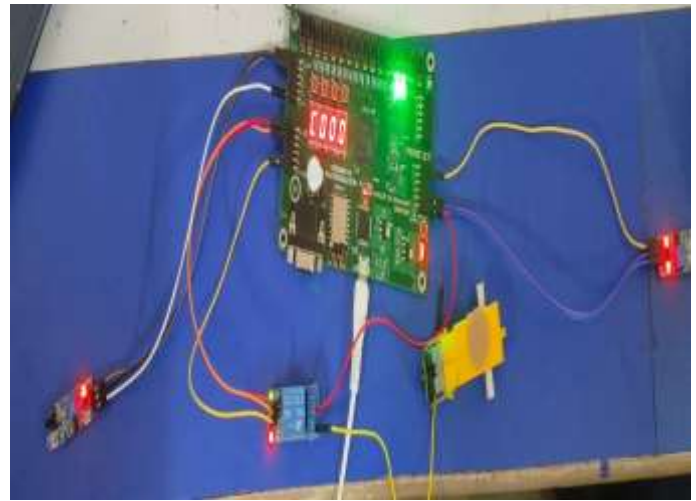
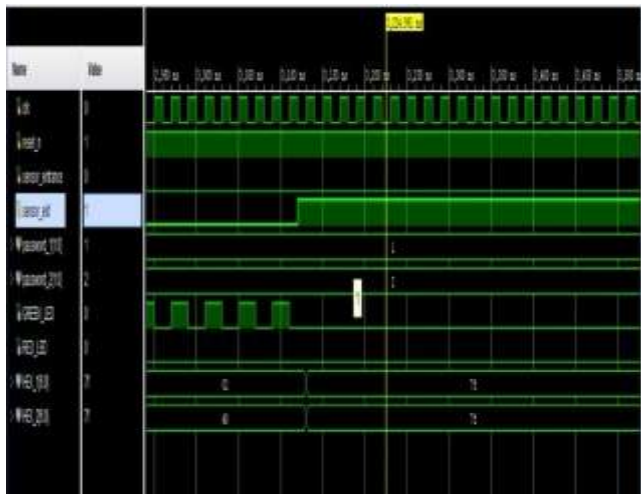


Fig. Simulation and Implementation result of car parking in parking exit with right password state

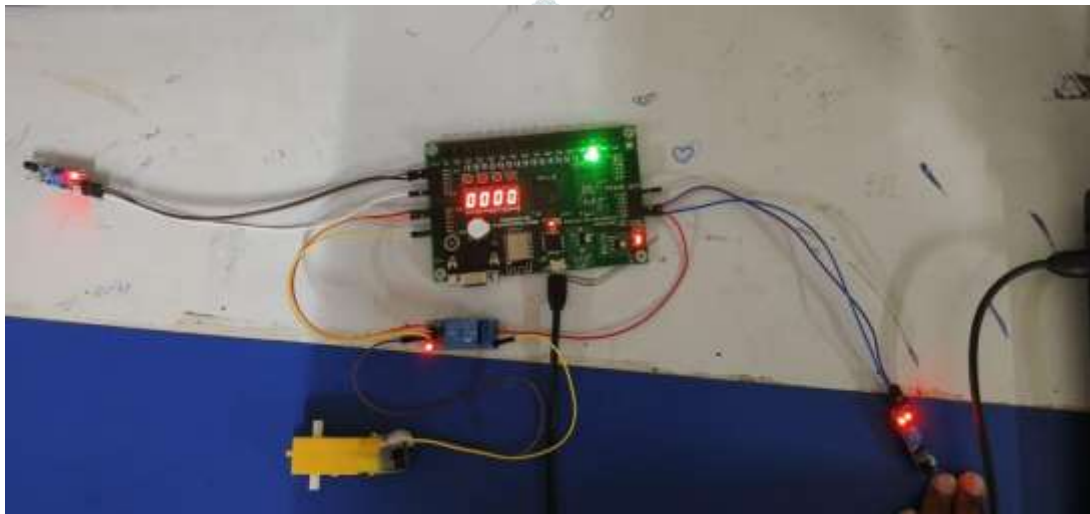


Fig. Implementation result in Parking exit state

V. CONCLUSION

An efficient car parking system that monitors vehicle entry and exist is successfully designed and implemented. The system caters to a wide range of issues. Device utilization was found to be much lower in comparison to the existing models. It is a safe and cost-effective personalized parking solution that focuses on the fundamental principle of saving time while parking. Because the majority of the operation is hardware based, maintaining the network of incoming and outgoing cars is also relatively simple. The concept also promotes quick parking. The proposed system would enable users to find car parking slots. If a driver is able to find a parking slot before reaching the area, it would definitely save a lot of time and energy. Implementation of this solution would help to reduce traffic conjunctions near parking areas and would help users to find slots without driving their vehicle all around the area. Which eventually saves the fuel and reduce pollution due to the toxic gases ejected from vehicle.

VI. REFERENCES

1. Trista Lin, Hervé Rivano, and Frédéric Le Mouël Improving parking availability prediction in smart cities with IoT and ensemble-based model. J. King Saud Univ. Comput. Inf. Sci. 2017
2. K. Y. Huang; S. B. Chang; P. R. Tsai. The advantage of the arduino sensing system on parking guidance information systems. Decis. Support Syst. 2016, 134, 113301
3. A. Z. M. Tahmidul Kabir; Nirmol Deb Nath; Fukrul Hasan; Rafin Akther Utshaw; Lovely Saha: Automated Parking System with Fee Management Using Arduino. Comput. Netw. 2015, 138, 18–30
4. Minal Patil; Vijay Chakole; Krushna Chetepawad. IoT Based Economic Smart Vehicle Parking System 114(7) p. 165-174 2014.